

AI/ML Engineer | Python | Data

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PROFESSIONAL SUMMARY:

- **Data & AI/ML Engineer** with **4+ years of experience** designing and deploying **AI/ML**, **Data Engineering**, and **Generative AI** solutions across **Healthcare**, **Banking**, **Insurance**, and **Telecommunications**.
- Experienced in developing **Machine Learning models**, **Natural Language Processing (NLP)** applications, **Retrieval-Augmented Generation (RAG)** pipelines, and **LLM-powered solutions** using **Python**, **LangChain**, **Hugging Face**, **FastAPI**, and **Vector Databases (FAISS, Pinecone)**.
- Strong background in designing and building scalable **Data Pipelines**, **ETL/ELT Workflows**, **Real-Time Data Processing**, and **MLOps** solutions using **Apache Spark**, **PySpark**, **Kafka**, **Airflow**, **Databricks**, **MLflow**, **Docker**, and **Kubernetes**.
- Hands-on experience deploying **Production-Ready AI/ML Applications** on **AWS**, **Azure**, and **GCP**, with expertise in **Model Deployment**, **Model Monitoring**, **CI/CD**, and **Cloud-Native Architectures**.
- Skilled in developing **RESTful APIs**, **Microservices**, and **Enterprise AI Solutions**, collaborating with **Cross-Functional Teams** to deliver **Scalable**, **Secure**, and **Business-Driven Applications**.
- Passionate about leveraging **Data Engineering**, **Machine Learning**, and **Generative AI** to build **Scalable Data Platforms**, **Intelligent Applications**, and **Business-Driven AI Solutions**.

TECHNICAL SKILLS:

Programming Languages	Python, SQL, Java, Scala, JavaScript, TypeScript
Generative AI & Artificial Intelligence	Generative AI, Large Language Models (LLMs), Natural Language Processing (NLP), Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, Vector Databases (FAISS, Pinecone)
Machine Learning	TensorFlow, PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, MLflow, Model Training, Model Deployment, Model Monitoring, Responsible AI
Data Engineering	Apache Spark, PySpark, Apache Kafka, Apache Airflow, Databricks, ETL/ELT Pipelines, Snowflake, Hadoop, Hive
Cloud & MLOps	AWS (SageMaker, S3), Azure AI, Azure OpenAI, Azure Machine Learning, GCP Vertex AI, Docker, Kubernetes, Kubeflow, Git, GitHub Actions, Azure DevOps, Jenkins
Backend Development	FastAPI, Flask, Spring Boot, RESTful APIs, Microservices
Databases	PostgreSQL, MySQL, MongoDB, BigQuery, DynamoDB

EDUCATION:

- **Master of Science in Computer Science**
Campbellsville University | Kentucky, USA - 2024
- **Bachelor of Technology in Computer Science**
Jawaharlal Nehru Technological University Kakinada | Andhra Pradesh, India - 2022

CERTIFICATIONS:

- AWS Certified Machine Learning Engineer - Amazon Web Services
- Google Cloud Certified - Generative AI Studio
- Machine Learning Engineering for Production Specialization - Coursera
- Generative AI for Developers - LinkedIn Learning
- GenAI Job Simulation - BCG X
- Data Science Job Simulation - BCG X

PROFESSIONAL EXPERIENCE:

Client: Walgreens | Chicago, IL

Nov 2025 – Present

Role: Generative AI Engineer

Responsibilities:

- Designed and optimized **Large Language Model (LLM)** solutions to automate **healthcare document summarization, patient communication, and pharmacy support workflows**.
- Built **Retrieval-Augmented Generation (RAG)** pipelines using **FAISS** and **Pinecone** to enable secure, context-aware retrieval from **healthcare knowledge bases, pharmacy documentation, and enterprise medical content**.
- Developed **Prompt Engineering Frameworks** with **Guardrails** to minimize **Hallucinations** and improve the **Accuracy, Consistency, and Reliability** of LLM-generated Responses.
- Processed and normalized **healthcare documents, prescription information, medical guidelines, and ICD-10/CPT data** to support AI model training and evaluation.
- Integrated AI-powered microservices with **enterprise healthcare applications and pharmacy management systems** using secure REST APIs while following **HIPAA-compliant** data handling practices.
- Improved **model performance** by optimizing latency, token usage, inference speed, and accuracy using domain-specific evaluation metrics.
- Deployed GenAI solutions with **FastAPI, Docker, Kubernetes**, and CI/CD pipelines for scalable production environments.
- Collaborated with **Business Stakeholders, Healthcare SMEs, Data Scientists, and Software Engineering Teams** to **Design, Validate, and Deploy AI Solutions** that improved **Healthcare and Pharmacy Workflows**.
- Created **medical document classification and entity extraction models** to identify diagnoses, medications, procedures, and clinical risks using **NLP + transformers**.
- Developed evaluation frameworks using **BLEU, ROUGE, Precision, Recall**, and **business-specific quality metrics** to improve the reliability and performance of Generative AI applications.

Environment: Python, FastAPI, PyTorch, Hugging Face, LangChain, Azure OpenAI, FAISS, Pinecone, SQL, Docker, Kubernetes, MLflow, Git, Azure DevOps (CI/CD), Azure.

Client: BNY | New York, NY

Dec 2024 – Oct 2025

Role: AI/ML Engineer

Responsibilities:

- Developed, tested, and deployed **Machine Learning Models** for **Fraud Detection, AML Monitoring, and Customer Segmentation**, improving risk analysis and operational efficiency.
- Built **NLP Pipelines** for **Document Processing, KYC Automation, Customer Communication Analysis, and Banking Document Extraction**.
- Implemented **Large Language Models (LLMs)** and **Retrieval-Augmented Generation (RAG)** solutions for **Intelligent Search, Document Retrieval, and Automated Decision Workflows**.
- Converted data science prototypes into **Production-Ready ML Services** using **Python, TensorFlow, PyTorch, Scikit-learn, and Spark ML**.
- Built and optimized **ETL/ELT Pipelines** for structured and unstructured banking data using **SQL, Apache Spark, Kafka, Databricks, and Airflow**.
- Designed and maintained **MLOps Pipelines** using **MLflow, Kubeflow, Azure ML, and AWS SageMaker** for **Model Training, Versioning, Deployment, and Monitoring**.
- Integrated **AI/ML Services** with enterprise banking applications through **REST APIs**, enabling scalable and secure AI-driven solutions.
- Developed **CI/CD Pipelines** using **Azure DevOps, Jenkins, and GitHub Actions** to automate model deployment, testing, and release management.
- Implemented **Model Monitoring** to detect **Data Drift, Model Drift, Bias**, and performance degradation, ensuring reliable production deployments.
- Prepared **Model Documentation, Validation Reports, Monitoring Dashboards, and Audit Artifacts**, while supporting **Responsible AI, Model Governance**, and regulatory compliance requirements.

Environment: Python, TensorFlow, PyTorch, Scikit-learn, Spark ML, SQL, Databricks, Kafka, Airflow, Azure ML, AWS SageMaker, MLflow, Kubeflow, Docker, Kubernetes, Azure OpenAI, Hugging Face, LangChain, LlamaIndex, Pinecone, FAISS, Git, Azure DevOps, Jenkins.

Client: Nationwide | Columbus, OH

Mar 2023 – Nov 2024

Role: Machine Learning Engineering

Responsibilities:

- Developed and deployed **Machine Learning Models** for **Fraud Detection**, **Claims Automation**, **Risk Assessment**, **Customer Segmentation**, and **Customer Retention**, improving insurance operations and decision-making.
- Designed and implemented end-to-end **Machine Learning Pipelines** for **Underwriting**, **Claims Processing**, **Policy Servicing**, and **Customer Support Automation**.
- Built **Natural Language Processing (NLP)** and **Retrieval-Augmented Generation (RAG)** solutions to process **Policy Documents**, **Claims Reports**, **FNOL (First Notice of Loss)** records, **Adjuster Notes**, and **Customer Communications**.
- Collaborated with **Actuarial Teams**, **Claims Analysts**, **Underwriters**, **Data Engineers**, and **Cloud Architects** to develop scalable AI/ML solutions aligned with business objectives.
- Improved **Model Accuracy**, **Inference Performance**, and **Scalability** using distributed computing techniques and machine learning optimization strategies.
- Implemented **MLOps Pipelines** for **Model Training**, **Versioning**, **Deployment**, **Monitoring**, and **Drift Detection** across **Development**, **UAT**, and **Production** environments.
- Integrated **Machine Learning Models** with enterprise **Policy Administration**, **Claims Management**, and **Customer Service Platforms** using secure **REST APIs**.
- Performed **Model Validation**, **A/B Testing**, and **Performance Evaluation** to ensure reliable model performance and business impact.
- Prepared **Technical Design Documents**, **Solution Architecture**, **Data Lineage Documentation**, and **Deployment Playbooks** to support enterprise AI/ML initiatives.
- Applied **Responsible AI**, **Data Privacy**, **Model Governance**, and **Insurance Compliance** standards throughout the AI/ML development lifecycle.

Environment: Python, TensorFlow, PyTorch, Scikit-learn, Azure AI, AWS SageMaker, GCP Vertex AI, Hugging Face, LangChain, LangGraph, MLflow, Docker, Kubernetes, Git, REST APIs.

Client: Wipro Ltd | Bangalore, India

Mar 2022 – Dec 2022

Role: AI/ML Python Developer

Responsibilities:

- Developed and maintained **Python-based Backend Applications** to automate telecom business processes and improve operational efficiency.
- Designed and implemented **RESTful APIs** using **FastAPI** for integration with **Oracle CRM**, **Billing Systems**, and enterprise applications.
- Built and optimized **ETL Pipelines** using **Python**, **Pandas**, and **SQL** to process customer, billing, and telecom operational data.
- Processed **Call Detail Records (CDRs)** and telecom usage data using **Pandas** and **NumPy** to support business analytics and reporting.
- Integrated Python services with **Oracle Service Bus (OSB)** and **Oracle SOA Suite** to enable seamless communication between enterprise applications.
- Developed reusable Python modules for consuming **REST** and **SOAP APIs**, improving application maintainability and reducing development effort.
- Developed **Machine Learning Models** using **Scikit-learn** and **XGBoost** to identify high-risk subscribers and support customer retention initiatives.
- Orchestrated **Data Ingestion** and scheduled batch processing workflows using **Apache Airflow**.
- Implemented **Apache Kafka Consumers** to process streaming telecom events for near real-time analytics.
- Containerized Python applications using **Docker** and supported **CI/CD Pipelines** using **Git**, **Jenkins**, and **Maven** for reliable deployments.
- Wrote optimized **SQL Queries**, performed **Data Validation**, and generated reconciliation reports to improve data quality and business reporting.
- Collaborated with **Oracle DBAs**, **QA Engineers**, **Business Analysts**, and cross-functional teams to troubleshoot production issues and optimize application performance.
- Applied **Secure Coding Practices**, **Data Masking**, **Logging**, and **Audit Controls** while handling customer and billing information.
- Participated in **Agile** ceremonies, including sprint planning, code reviews, testing, production deployments, and application support.

Environment: Python, FastAPI, Pandas, NumPy, SQL, Scikit-learn, XGBoost, Apache Airflow, Apache Kafka, Oracle BRM, Oracle Siebel CRM, Oracle Service Bus (OSB), Oracle SOA Suite, REST APIs, SOAP/WSDL, Docker, Git, Jenkins, Maven, SonarQube, Jupyter Notebook.

Client: BSNL Ltd | Hyderabad, India

July 2021 – Feb 2022

Role: Data Engineer

Responsibilities:

- created and oversaw ETL/ELT pipelines to facilitate reporting and analytics for both structured and unstructured data.
- Designed, developed, and maintained **ETL/ELT Pipelines** to ingest, transform, and load **Structured** and **Unstructured Data** from multiple enterprise data sources.
- Built scalable **Data Ingestion Pipelines** using **Apache Spark**, **PySpark**, **Kafka**, and **Apache Airflow** to support batch and real-time data processing.
- Collaborated with **Data Scientists**, **Business Analysts**, and **Software Engineers** to deliver high-quality datasets for **Analytics**, **Reporting**, and **AI/ML Workloads**.
- Optimized **SQL Queries**, **Data Transformations**, and **Data Pipelines** to improve processing performance, scalability, and data reliability.
- Developed and maintained **Data Models**, **Database Schemas**, and **Relational/NoSQL Databases** including **PostgreSQL**, **MySQL**, and **MongoDB**.
- Implemented **Data Validation**, **Data Quality Checks**, and **Pipeline Monitoring** to ensure data accuracy, consistency, and completeness.
- Supported cloud-based data processing by working with **AWS** services for scalable storage and compute resources.
- Implemented **Data Governance**, **Role-Based Access Control (RBAC)**, and **Data Security** practices to protect enterprise data assets.
- Prepared **Technical Documentation**, **Pipeline Architecture Diagrams**, and **Operational Runbooks** to support production deployments and maintenance.
- Supported **Business Intelligence**, **Reporting**, and **Machine Learning Teams** by delivering reliable and well-structured datasets for downstream applications.

Environment: Python, SQL, Apache Spark, PySpark, Apache Kafka, Apache Airflow, PostgreSQL, MySQL, MongoDB, Hadoop, AWS, Git, Docker.

